

Traffic Flow Detection and Adaptive Signal Control Using CCTV Imagery

¹Arpita Bhandari, ²Dev Dandekar, ³Hariom Gilda, ⁴Prem Rahinj & ⁵Dr. Archana Ghotkar

¹Student;SCTR's Pune Institute of Computer Technology (C.E.), Pune, Maharashtra, India, arpitabhandari2511@gmail.com

²Student;SCTR's Pune Institute of Computer Technology (C.E.), Pune, Maharashtra, India, devdandekar365@gmail.com

³Student;SCTR's Pune Institute of Computer Technology (C.E.), Pune, Maharashtra, India, gildahariom@gmail.com

⁴Student; SCTR's Pune Institute of Computer Technology (C.E.), Pune, Maharashtra, India, premrahinj73@gmail.com

⁵Associate Professor; SCTR's Pune Institute of Computer Technology (C.E.), Pune, Maharashtra, India, aaghotkar@pict.edu

Abstract

Traffic saturation and congestion in urban areas have become an issue in rapidly growing regions, primarily due to the continued reliance on rigid, fixed-schedule traffic signal systems that are unable to respond to real-time variations in vehicle flow. As urban populations increase and vehicle density rises, these static systems lead to inefficient traffic management, longer waiting times, increased fuel consumption, and higher emissions. Although modern solutions have introduced sensors and CCTV-based monitoring, most existing approaches operate in isolation, focusing on optimizing individual intersections rather than considering the broader traffic network. This lack of coordination often results in bottlenecks shifting from one junction to another, rather than being effectively resolved. To report these restrictions, this paper suggests an AI-driven traffic management system with ability of leveraging CCTV imagery to dynamically interpret traffic conditions. By utilizing YOLOv11 for vehicle detection (achieving $mAP@50 > 0.91$) and ByteTrack for multi-object tracking, the system computes density-per-lane, vehicle-queue-length, and prioritized vehicle presence in real time. The BFS(Breadth First Search)-coordinated adaptive signal control algorithm, evaluated across a six-junction SUMO network with 810 vehicles, achieves a striking average wait time reduction ($58.41\text{ s} \rightarrow 37.12\text{ s}$; about 36%), a time loss reduction by 30.87% ($78.09\text{ s} \rightarrow 53.98\text{ s}$), an average trip duration reduction of 19.50%, and about 70% reduction in priority vehicle waiting time ($51.88\text{ s} \rightarrow 17.71\text{ s}$) compared to a fixed-cycle baseline. Average vehicle speed improves from 5.3 m/s to 6.8 m/s. These results demonstrate the system's precision, recall, and coordination efficiency across heterogeneous urban traffic.

Keywords: Adaptive traffic signal control, CCTV imagery, deep learning, vehicle detection, YOLO, ByteTrack, multi-intersection coordination

1. Introduction

Congestion of traffic has become a persistent urban environment challenge due to the rapid growth of vehicle ownership and limited expansion of road infrastructure [1],[2]. The imbalance between traffic demand and road capacity often results in prolonged travel time, higher fuel consumption, and increased environmental pollution. These problems not only reduce transportation efficiency but also have social and economic consequences for city residents and governing authorities.

Conventional traffic control systems generally operate using fixed-time signal cycles designed from historical traffic data [3]. Although such systems are simple to implement, they cannot respond to real-time variations in vehicle flow, road incidents, or peak-hour surges. This lack of adaptability causes inefficient signal distribution, excessive queue formation, and unnecessary idling at intersections. In densely populated areas, these inefficiencies accumulate, amplifying congestion across connected junctions.

To improve road network performance, modern research focuses on adaptive traffic signal control systems that adjust light durations dynamically according to current traffic density. Earlier adaptive models relied on sensors such as inductive loops and radar detectors to count or detect vehicles. However, these hardware-based solutions are expensive to deploy and cover only a limited detection area.

The widespread installation of closed-circuit television (CCTV) cameras and the advancement of computer vision and artificial intelligence (AI) have created new possibilities for intelligent traffic management. By analyzing real-time video feeds, vision-based systems can automatically identify, classify, and track vehicles without additional roadside equipment. Algorithms like You Only Look Once (YOLO) for object detection [5] and ByteTrack for multi-object

tracking [12] enable reliable traffic analysis in complex scenes and varying weather conditions. These systems can estimate traffic density, queue length, and flow direction, which form the basis for adaptive signal timing.

Despite significant progress, most current systems focus on controlling single intersections independently. The absence of coordination among multiple signals along a corridor leads to stop-and-go movement, fuel wastage, and increased delays. Recent research aims to overcome this limitation through network-level adaptive control, where intersections are synchronized using graph-based decision algorithms or reinforcement learning to achieve smoother traffic progression across multiple junctions.

This survey reviews the evolution of traffic control mechanisms from traditional static models to advanced, vision-based adaptive systems. It summarizes the methods and technologies used in recent studies, identifies persistent limitations, and highlights opportunities for developing scalable and cooperative AI-driven traffic management frameworks that can support the smart cities of the future.

Literature Survey

Over the previous rare periods, traffic management and control schemes have undergone a gradual evolution from static, pre-timed signal systems to adaptive, sensor-based, and now intelligent vision-driven systems. This section reviews the major approaches, highlighting their principles, benefits, and limitations.

1.1 Conventional Traffic Control Systems

Traditional traffic control workflows rely on fixed interval cycles determined from historical or empirical data. Models such as Webster's method [1] and the Indian Roads Congress (IRC) standards [2] were developed to optimize signal phases for average flow conditions. These systems assume predictable traffic behavior, where signal timings remain constant within preset intervals throughout the day.

While these models achieved acceptable performance in less congested networks, they proved inefficient in dynamic urban environments where traffic volumes vary significantly during different hours. The study by researchers in [1] observed that such systems do not respond to real-time changes caused by weather conditions, road accidents, or special events. Moreover, they often fail to accommodate sudden fluctuations in vehicle density, leading to congestion during peak hours and underutilization of green time during off-peak periods.

Another major limitation is the lack of synchronization among multiple intersections. Most fixed-time systems treat each signal as an isolated unit, which disrupts vehicle progression across a network of intersections. Consequently, vehicles are forced to stop repeatedly, increasing idling, fuel consumption, and emissions. Studies comparing IRC guidelines with adaptive alternatives [2] concluded that traditional systems lack scalability and are unsustainable for large metropolitan networks.

1.2 Sensor-Based Adaptive Systems

To address the limitations of fixed-timing mechanisms, researchers began exploring actuated and sensor-based systems capable of responding to real-time conditions [3]. These systems use sensors such as inductive loops, infrared detectors, and radar sensors to monitor vehicle presence, density, and flow direction. When the sensor detects increased traffic in one lane, the controller automatically extends the green phase duration to balance the load.

Although these systems improved intersection efficiency to some degree, they introduced several new challenges. As discussed in [3], inductive loops are costly to install because they require cutting the road surface, making maintenance difficult and time-consuming. Similarly, infrared and radar sensors can malfunction during rain, fog, or extreme heat. Furthermore, these systems primarily detect vehicle presence but cannot distinguish between vehicle types (e.g., cars, buses, or trucks), limiting their analytical usefulness.

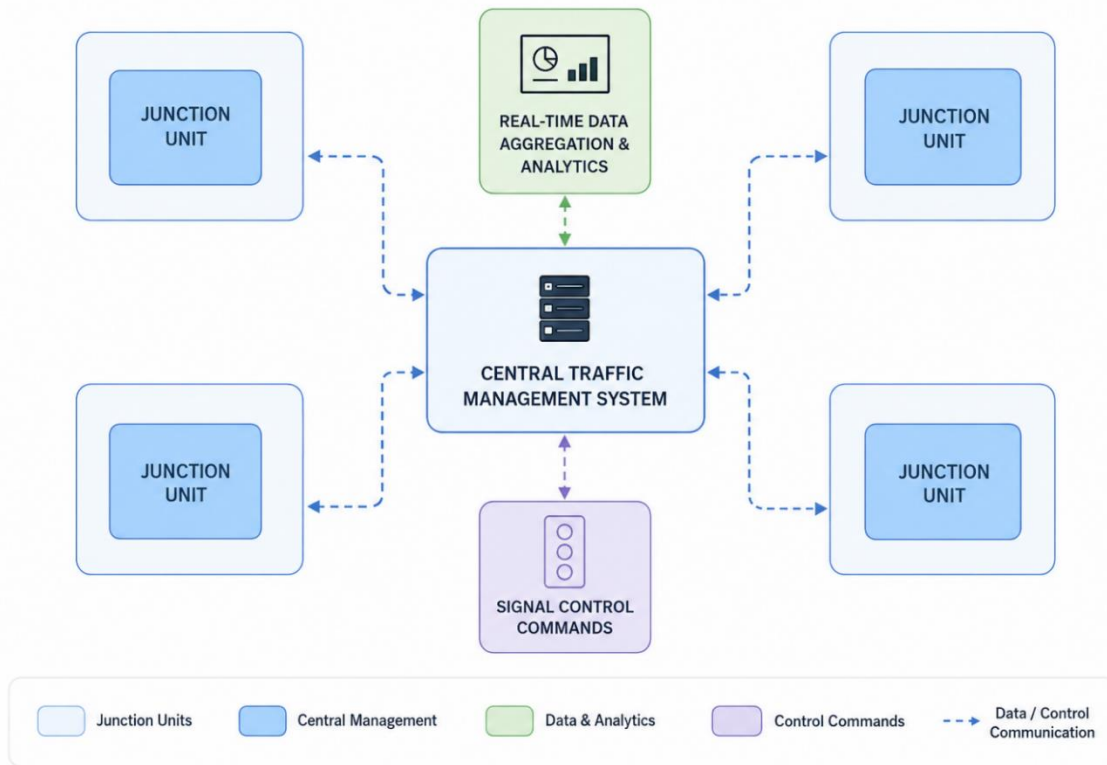


Fig.1. Inductive loops-based Architecture for efficient dynamic traffic control system

Another issue is the fragmented nature of control. Even with sensors, many adaptive systems still operate at a single intersection level without communication between adjacent signals. This partial adaptability often transfers congestion from one junction to the next rather than solving it holistically.

1.3 Wireless and IoT-Based Traffic Monitoring

With the evolutions in WSN and the IoT domains, research has shifted toward distributed systems that connect multiple intersections through networked data exchange. In [4], authors proposed a WSN-based adaptive traffic control model that collects data from road-side sensor nodes to compute vehicle density and average waiting time. The results showed improvement in local intersection performance compared to isolated systems.

However, WSN-based models still rely heavily on hardware installations and are constrained by energy consumption, communication delay, and data loss during transmission [4]. The limited coverage of each sensor node means these systems are best suited for small-scale deployments. Additionally, integration with legacy traffic controllers remains a significant technical challenge.

IoT-based frameworks extend these concepts by enabling real-time communication between sensors, controllers, and cloud servers. Though effective in data collection and coordination, such systems involve high infrastructure costs and are vulnerable to network failures and latency issues in busy city environments.

1.4 Vision-Based Traffic Detection and Control

The increasing availability of CCTV infrastructure has transformed the field of intelligent transportation systems (ITS). Vision-based techniques use camera feeds to analyze live traffic conditions without relying on ground-embedded or active sensors. In contrast to loop detectors, cameras can provide richer contextual information such as vehicle type, motion trajectory, and queue length.

Early computer vision approaches used techniques like background subtraction and frame differencing to detect moving vehicles. However, these classical methods were sensitive to shadows, lighting variations, and occlusions. The introduction of DL-based object detection models revolutionized the speed as well as accuracy of vehicle

recognition. Algorithms such as YOLO, Faster R-CNN, and SSD demonstrated strong performance in identifying and classifying vehicles from real-time CCTV feeds [5].

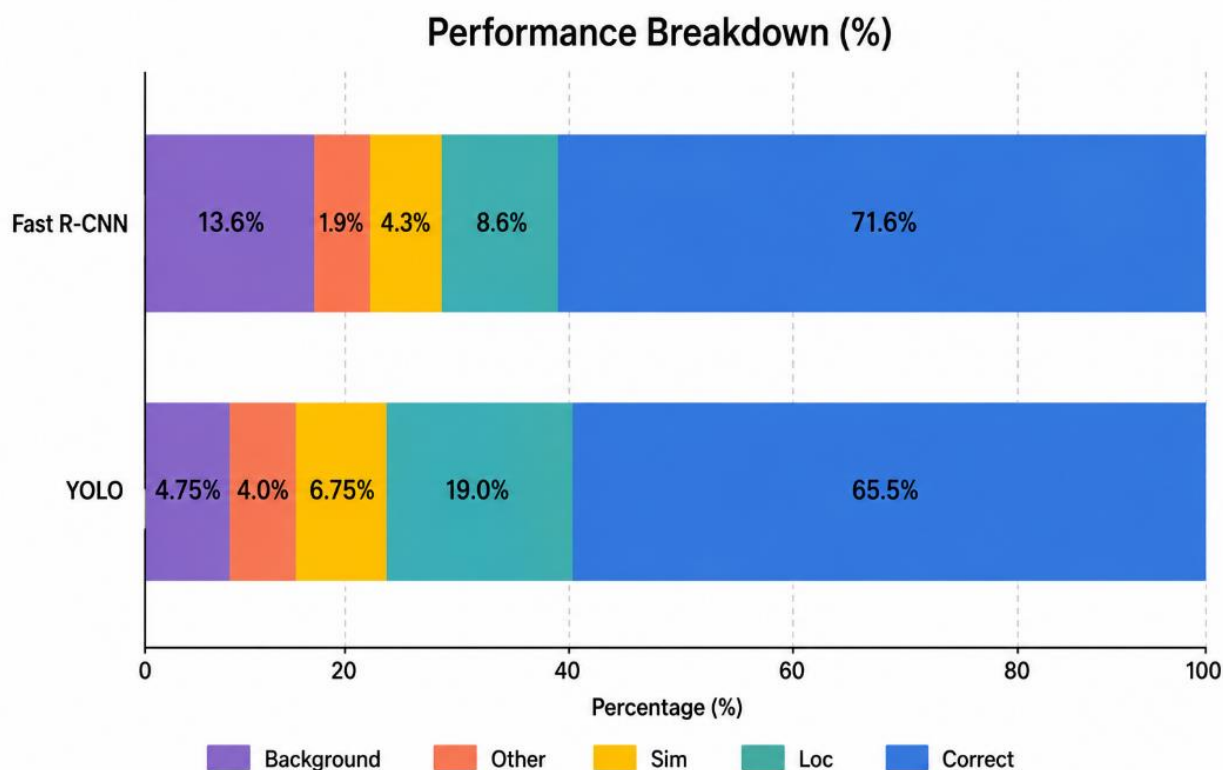


Fig.2. Error Analysis by Fast R-CNN & YOLO

Recent research emphasizes the integration of object detection with multi-object tracking algorithms such as ByteTrack, enabling continuous tracking of vehicles across frames. This helps in estimating real-time traffic density, queue lengths, and lane-wise occupancy [12]. This combination of YOLO for object detection and ByteTrack for tracking has shown excellent results in complex, multi-lane environments.

Despite these advancements, vision-based systems face challenges like occlusions during heavy congestion, low visibility during nighttime, and computational demands for real-time processing. Researchers are now investigating optimized architectures, edge computing, and lightweight models to enhance real-time inference capabilities.

1.5 Adaptive and Coordinated Signal Control

Most of the earlier adaptive systems focused on optimizing individual intersections, resulting in local improvements but limited network-wide benefits. Studies such as [6] and [7] demonstrated that coordinated control, where multiple intersections share real-time data, can significantly reduce overall travel time and improve fuel efficiency. These models often use graph-based decision algorithms such as the Weighted Mean Phase Analysis (WMPA) or reinforcement learning-based control to determine optimal signal phases.

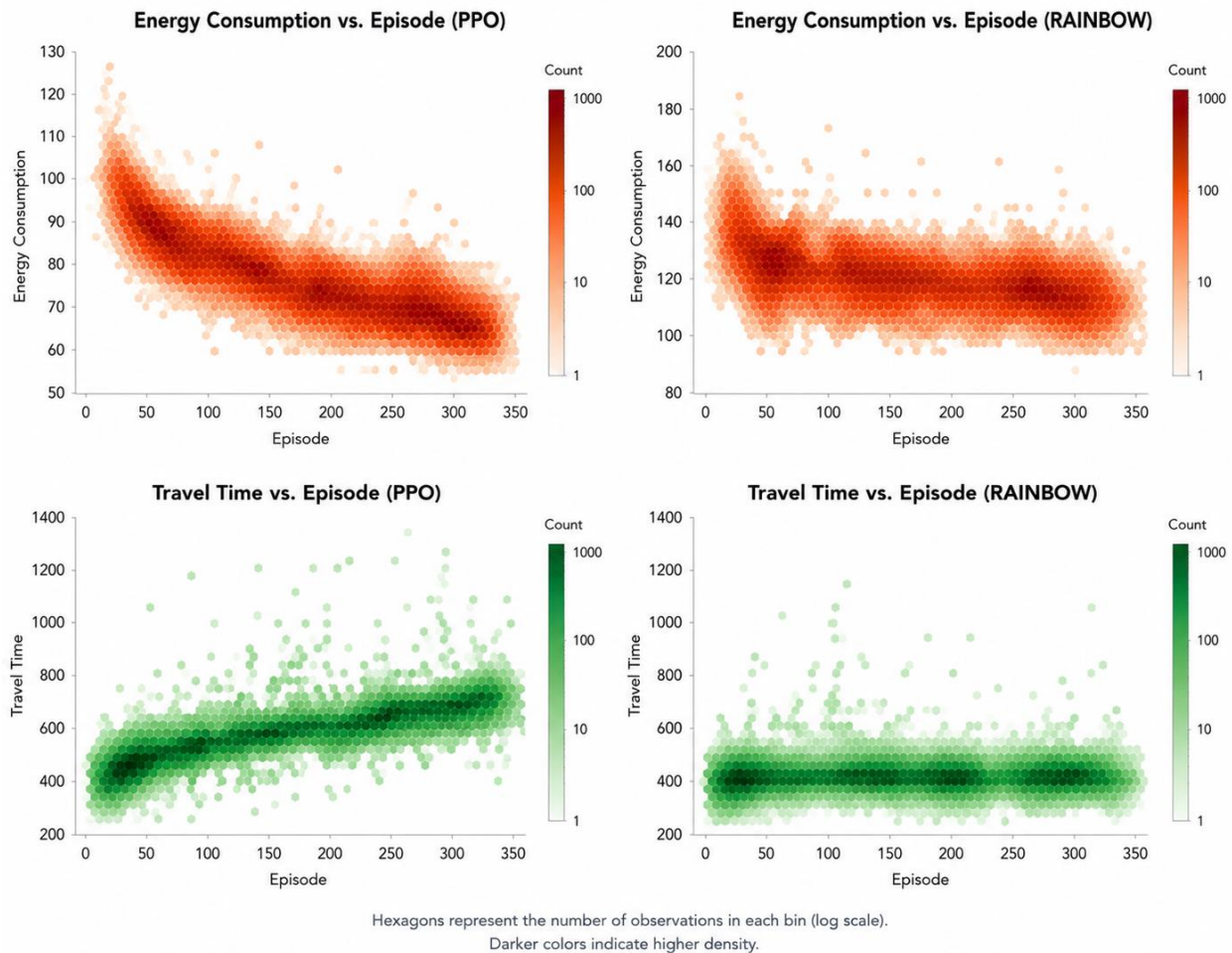


Fig.3. Analysis of key matrices in the training processes of PPO and RAINBOW algorithms.

However, real-time coordination across several junctions introduces its own set of challenges, including high communication overhead, synchronization delays, and increased algorithmic complexity. Moreover, deploying such intelligent systems in developing countries requires cost-effective infrastructure, robust connectivity, and adaptable frameworks compatible with existing signal hardware.

1.6 Summary of Observations

From the reviewed literature, it is evident that while sensor-based and IoT-driven systems laid the groundwork for real-time adaptivity, vision-based AI models have become the most promising direction for the future of intelligent traffic control. These models not only reduce dependency on physical hardware but also provide granular insights into vehicle behavior and traffic dynamics. Nevertheless, a fully coordinated and scalable solution that integrates vision analytics with real-time adaptive control remains an open research challenge.

2. Proposed Methods

The development of intelligent traffic control systems has been strongly influenced by progress in computer vision, machine learning, and optimization algorithms. These methodologies collectively aim to enhance real-time traffic responsiveness, minimize congestion, and improve overall network efficiency. This section reviews the key techniques studied for building adaptive, vision-based traffic systems.

2.1 Real-Time Vehicle Detection Using Deep Learning (YOLO)

Vehicle detection represents the first and most crucial step in vision-based traffic monitoring. Conventional detection algorithms, such as background subtraction and frame differencing, often failed under dynamic environmental

conditions like poor illumination, occlusion, or shadow interference. The introduction of deep learning models, particularly the LO family, is transformational for dynamic object detection through its speed and accuracy.

The YOLO framework divides an image into a grid and concurrently forecasts bounding boxes and class likelihoods [19], performing detection in a single pass through the network. This single-shot approach allows extremely high inference speeds while maintaining robust accuracy [8]. The comprehensive review in [9] discusses the progressive improvements from YOLOv1 to YOLOv8, including better backbone architectures, multi-scale feature fusion, and anchor-free detection heads. In traffic management applications, the YOLO family of object detection models has demonstrated strong capability in accurately classifying multiple vehicle categories in real-time scenarios. The YOLOv4 model introduces several architectural improvements and data augmentation strategies that significantly enhance detection performance and generalization ability across diverse visual conditions [10]. Building upon these advancements, subsequent versions of YOLO further optimize the trade-off between detection accuracy & inference speed. The comprehensive evolution of the YOLO framework, as summarized in [11], highlights suitability for deployment in edge-based intelligent transportation systems, where low latency, computational efficiency, and reliable detection are critical for adaptive traffic control.

Thus, YOLO-based detection serves as the foundation for higher-level modules, providing continuous real-time data on vehicle count, type, and positioning, which drive signal optimization decisions.

2.2 Multi-Object Tracking (ByteTrack)

Following detection, tracking individual vehicles across successive frames enables deeper insights into vehicle trajectories, flow direction, and queue formation. Among multi-object tracking algorithms, ByteTrack has emerged as a preferred choice due to its superior balance between accurateness and computational efficacy.

ByteTrack associates detected bounding boxes frame-by-frame using intersection-over-union (IoU) and appearance similarity scores. The method is designed to handle missed detections and identity switches effectively, which are common challenges in crowded intersections. As demonstrated in [12], ByteTrack achieves significantly higher IDF1 values compared to older methods like DeepSORT.

Further, studies integrating YOLO with ByteTrack confirm the combined pipeline's ability to accurately track and count vehicles under real-world conditions [13]. This hybrid detection-tracking approach enables reliable measurement of traffic flow parameters essential for adaptive signal adjustment.

2.3 Traffic Density and Queue Estimation

Estimating vehicle density and queue length at intersections is critical for determining dynamic signal durations. Earlier systems depended on inductive loops or infrared sensors, but these lacked the ability to differentiate between vehicle types or accurately capture queue buildup. Vision-based methods now use temporal video analysis combined with detection and tracking data to estimate congestion in real time.

Incorporating YOLO detections with temporal smoothing and motion analysis helps generate accurate queue length predictions even in partially occluded scenes. Studies have explored the use of CNNs to quantify traffic density directly from CCTV imagery [14]. The output from these systems provides a continuous numerical representation of congestion, forming the input to adaptive control algorithms.

2.4 Adaptive Signal Control Using Graph-Based Optimization

Dynamic traffic signal optimization aims to allocate green light durations proportionally to real-time traffic demand. Graph-based models provide an efficient mathematical structure for this task. The Weighted Mean Phase Adjustment (WMPA) algorithm treats every intersection as a node in a weighted graph, where the weights correspond to the measured traffic densities and queue lengths.

Research has shown that WMPA minimizes average waiting time by iteratively adjusting signal states based on observed conditions [15]. It ensures that vehicles approaching from heavily congested lanes receive longer green intervals, while maintaining coordination across neighboring intersections.

The inclusion of feedback loops and reinforcement learning into these graph-based approaches further enhances adaptability under unpredictable scenarios such as accidents or sudden surges in vehicle flow [16]. This integration marks a shift toward self-learning traffic control mechanisms capable of city-wide optimization.

It has been seen that the adaptive reinforcement learning models outperform the fixed-control methods in addressing traffic congestion problems [16]. From all the models used, G-DQN proves to be the best performer as it keeps the

queue lengths lowest during the learning process. This model is seen to perform effectively as a result of gradual learning and adaptation to traffic situations. Another model, A2C, proves to be another good performer by demonstrating an ability to reduce congestion levels gradually over time. On the other hand, DQN proves to have higher queue lengths compared to others, and hence less effective in terms of optimization and control over time. As far as the fixed-control model is concerned, it proved to be least efficient as it always has higher queue lengths.

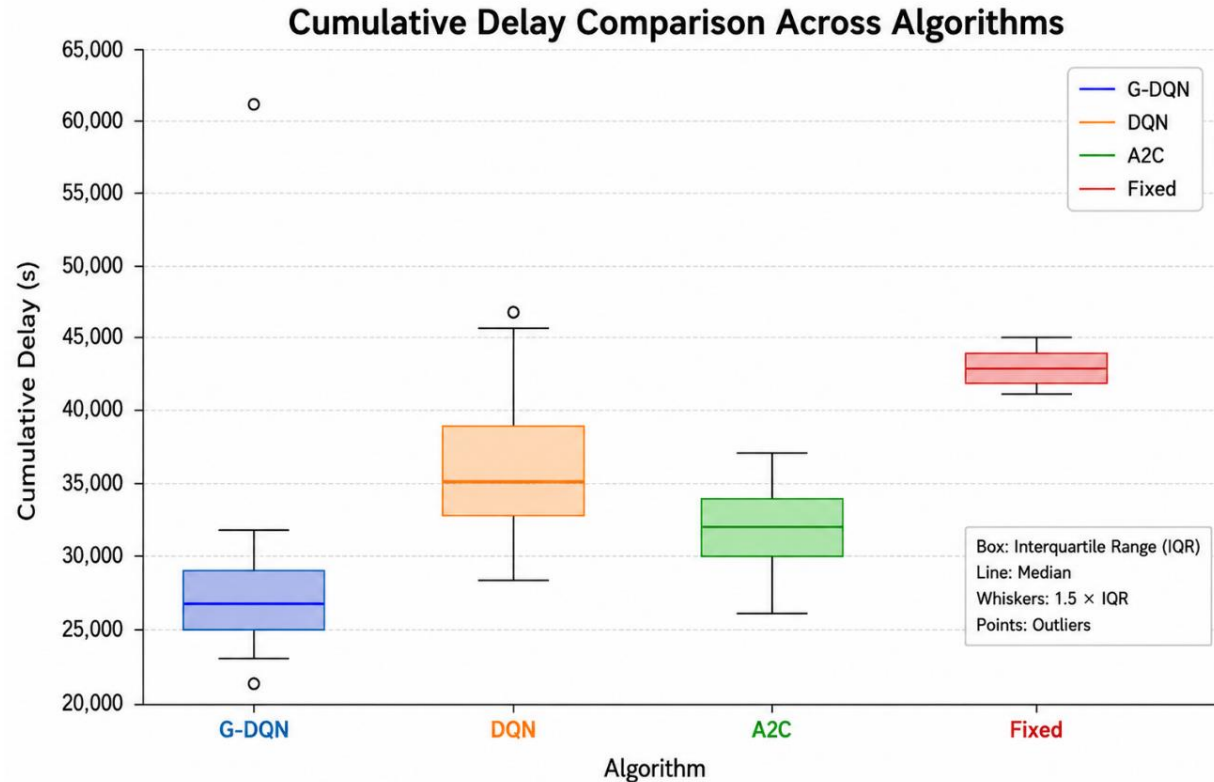


Fig.4. Comparison of cumulative queuing time of the four methods.

2.5 Multi-Intersection Coordination

Most existing adaptive systems operate on single intersections. However, achieving overall urban efficiency requires synchronization across multiple junctions. Research has highlighted the performance benefits of multi-junction coordination, showing reductions in CO₂ emissions due to smoother progression [6].

Recent methods use distributed control architectures where each intersection functions as an intelligent agent, exchanging summarized traffic data with neighboring nodes. Graph-based optimization and consensus algorithms are increasingly used to achieve real-time coordination while minimizing latency. Such approaches prevent bottleneck formation and ensure consistent vehicle progression across the network.

2.6 Simulation Models for Evaluation

Given the limited availability of suited datasets, simulation tools are essential for testing adaptive algorithms before real-world deployment. The SUMO (Simulation of Urban Mobility) and VISSIM platforms are widely used to replicate traffic conditions and evaluate new control strategies.

Researchers have demonstrated that SUMO provides realistic modeling of intersection-level dynamics and allows easy integration of AI-based controllers [17]. Similarly, studies have evaluated reinforcement learning and evolutionary optimization methods within simulated urban networks, achieving significant reductions in average waiting time [18].

These simulation environments enable safe, cost-effective experimentation and serve as a bridge between theoretical model design and real-world implementation.

2.7 Summary

The methodologies explored-YOLO-based detection, ByteTrack tracking, density estimation, graph-based optimization, and simulation validation-collectively represent the backbone of current research in adaptive traffic control. Their integration enables an intelligent ecosystem where CCTV imagery, deep learning, and optimization work together to enhance traffic efficiency, reduce congestion, and promote sustainable urban mobility.

3. System Architecture

The system consists of four layers: the Perception Layer, which captures CCTV video feeds and performs real-time vehicle detection and tracking using YOLO and ByteTrack; the Analytics Layer, which evaluates traffic density, queue length, and identifies alternative vehicles; the Control Layer, which applies BFS-based coordinated signal control and adaptive decision-making for traffic prioritization; and the Network Layer, which represents the six-junction SUMO-based urban traffic network where the generated signal decisions are executed and continuously updated through feedback.

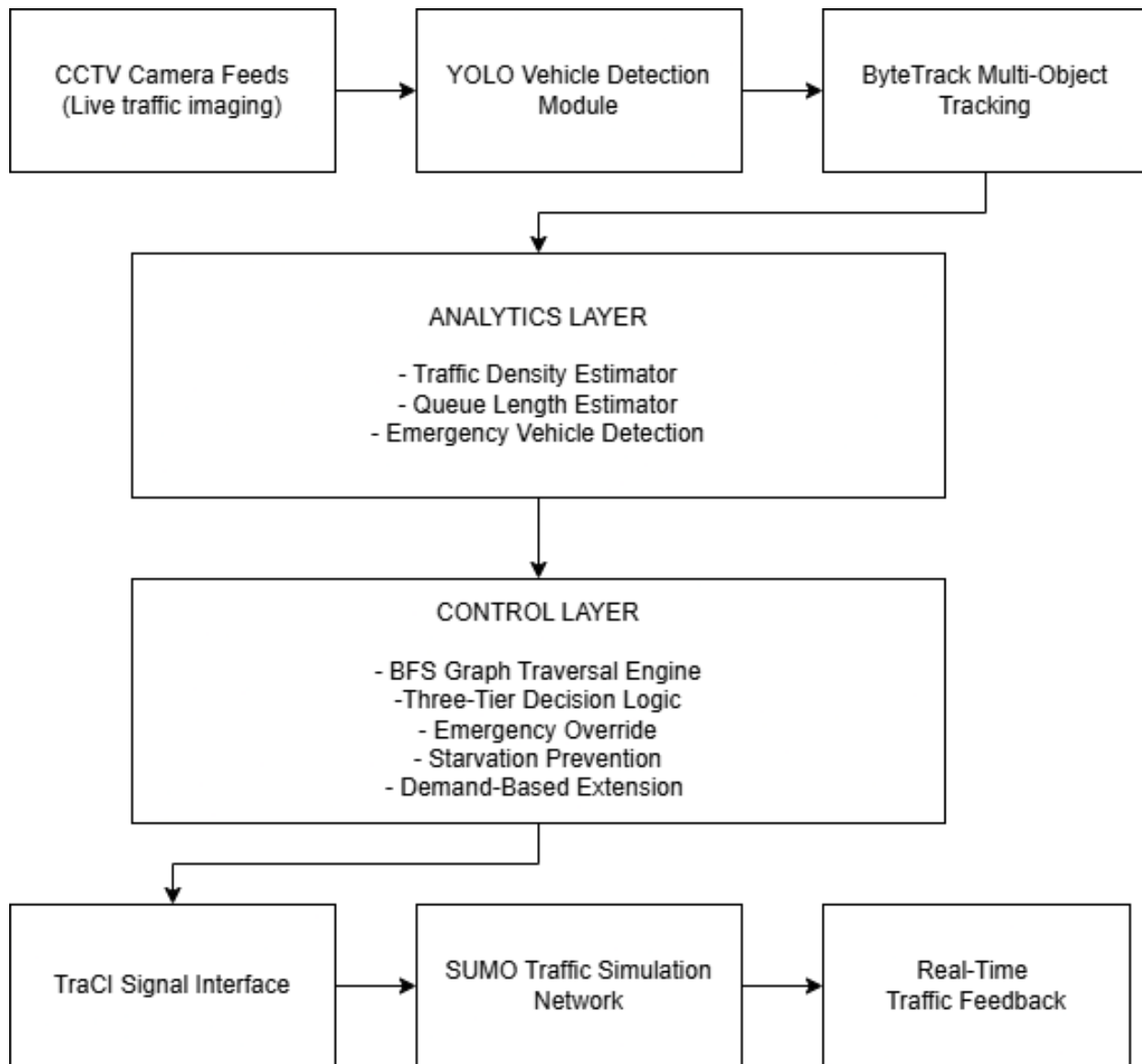


Fig.5. System architecture of the proposed Adaptive Traffic Signal Control System

3.1 Methodologies and Algorithms

3.1.1 YOLOv11 Vehicle Detection: Real-time vehicle detection is performed using YOLO, which identifies vehicle classes and generates per-edge traffic counts for adaptive signal control [5].

3.1.2 ByteTrack Multi-Object Tracking: ByteTrack associates detected vehicles across frames using IoU-based matching, enabling reliable trajectory tracking and queue estimation even in dense traffic [12].

3.1.3 BFS-Coordinated Control: The traffic network is modeled as a weighted graph where junction priorities are propagated using BFS from the highest-demand intersection to coordinate green signal progression.

ALGORITHM 1(*Pseudocode*):

Algorithm 1: BFS Priority Propagation

Input:

Junction Graph $G(V, E)$

Traffic Demand $w(v)$

Output:

Priority Phase Map P

```

1. root ← junction with maximum traffic demand
2. Initialize empty queue Q
3. Mark root as visited
4. Push root into Q
5. while Q is not empty do
6.   current ← dequeue(Q)
7.   for each neighboring junction n of current do
8.     if n is not visited then
9.       Assign priority phase based on incoming traffic
10.      Store priority in  $P[n]$ 
11.      Mark n as visited
12.      enqueue(Q, n)
13.     end if
14.   end for
15. end while
16. return P

```

3.1.4 Three-Tier Decision Logic: Signal control decisions follow three priority levels:

- Emergency vehicle override
- Starvation prevention
- Demand-based green signal extension ensuring adaptive and fair traffic management.

Algorithm 2: Three-Tier Signal Decision Logic

Input:

Current Junction j

Traffic Demand Data

Priority Phase Map P

```

1. for each junction j do
2.   // Tier 1: Emergency Override
3.   if emergency vehicle detected then
4.     Activate emergency phase
5.     Extend green duration

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6.     continue
7.     end if
8.     // Tier 2: Starvation Prevention
9.     for each inactive phase p do
10.        if waiting time of p exceeds threshold then
11.            Switch signal to phase p
12.        end if
13.    end for
14.    // Tier 3: Demand-Based Extension
15.    if current phase demand is high then
16.        Extend green signal duration
17.    else if current phase demand is zero then
18.        Skip current phase
19.    end if
20. end for

```

3.1.5 Decentralized Junction Coordination: Each junction independently executes control logic while periodically receiving BFS priority updates, improving scalability and fault tolerance.

3.1.6 Green Wave Corridor Scheduling: Predefined corridor offsets are integrated with BFS priority assignment to create smoother vehicle progression along major traffic routes.

4. Results & Discussion

4.1 Challenges

Though several progressions have been made in adaptive and vision-based traffic control, several challenges continue to limit their scalability and real-world effectiveness. These issues arise from environmental factors, infrastructure constraints, data limitations, and ethical considerations that collectively hinder the deployment of reliable, city-wide intelligent systems.

4.1.1 Environmental and Lighting Variations

Vision-based systems depend heavily on clear visual data, which is affected by weather, lighting, and shadows. Rain, fog, or night-time conditions often reduce image quality and detection accuracy. While newer deep learning models have improved robustness, performance inconsistencies still occur under poor visibility. Deploying supplementary infrared or thermal imaging technologies can help but increases installation and maintenance costs.

4.1.2 Occlusion and High-Density Traffic

At crowded intersections, overlapping vehicles and irregular traffic patterns make it difficult for tracking algorithms to maintain identity consistency. In dense urban conditions, even advanced models experience ID switches and false detections, affecting vehicle counting and congestion estimation. Such inaccuracies impact the reliability of adaptive signal decisions.

4.1.3 Dataset and Generalization Issues

Most detection models are trained on structured datasets that fail to represent the diverse, mixed traffic seen in many cities. As a result, models often perform poorly in unstructured conditions with varied vehicle types. Building large, annotated, region-specific datasets remain a significant research and resource challenge.

4.1.4 Computational and Infrastructure Constraints

Real-time processing of CCTV feeds demands high computational power and stable connectivity between intersections. Many cities still operate outdated control units incapable of supporting deep learning models. Limited bandwidth and latency issues further restrict real-time coordination across multiple junctions.

4.1.5 Privacy and Ethical Concerns

Since CCTV footage can capture identifiable information such as faces or license plates, data privacy becomes a key issue. The lack of strong privacy-preserving mechanisms and standardized governance frameworks raises concerns

about surveillance misuse. Ensuring ethical deployment requires anonymization, encryption, and localized edge processing.

4.1.6 Maintenance and Reliability

Outdoor cameras are exposed to dust, misalignment, and environmental wear, which degrade performance over time. Irregular maintenance or system downtime disrupts adaptive control mechanisms, emphasizing the need for sustainable upkeep and monitoring frameworks.

4.1.7 Cost and Scalability

Implementing adaptive control systems across large cities requires significant investment in hardware, network infrastructure, and integration. While small-scale experiments show promise, scaling to multi-junction coordination remains complex due to interoperability and cost constraints.

4.2 Future Scope

The future of vision-based adaptive traffic management lies in developing more intelligent, responsive, and inclusive systems that not only optimize vehicle flow but also address the dynamic realities of urban mobility. As urban populations grow and transportation systems become increasingly complex, next-generation solutions must focus on broader data integration, fairness, safety, and adaptability.

4.2.1 Integration of Multimodal Data Sources

Future advancements should move beyond reliance on CCTV feeds alone. Integrating data from additional sources such as radar sensors, GPS signals, IoT-based vehicle detectors, and connected vehicle networks can significantly improve the accuracy of traffic analysis. This multimodal approach will ensure that system decisions remain reliable even under poor visibility, adverse weather, or sensor failures. Such integration will also help detect pedestrian and cyclist movement, improving overall road safety and inclusivity.

4.2.2 Priority Management for Emergency and Special Vehicles

An essential enhancement for adaptive traffic systems involves automated priority handling for priority automobiles such as ambulances, fire engines, and police cars. By integrating vehicle-mounted transmitters or AI-based visual recognition, the system can detect approaching emergency vehicles and adjust signals dynamically to create a “green corridor.” This feature not only reduces emergency response times but also demonstrates the potential of intelligent systems in life-saving applications. Implementing real-time priority control in coordination with hospitals, fire stations, and law enforcement agencies could make urban traffic significantly safer and more efficient.

4.2.3 Creation of Region-Specific and Realistic Datasets

The development of large-scale, annotated, and diverse datasets representing local traffic patterns is vital for model generalization. Most existing datasets are collected from structured environments that do not accurately depict the heterogeneous and unorganized traffic common in many developing regions. Future work should emphasize building datasets that capture various weather conditions, lighting variations, and mixed vehicle types. Establishing public, standardized datasets will also help researchers evaluate model performance under comparable conditions and accelerate innovation in traffic analysis algorithms.

4.2.4 Collaborative and Coordinated Intersection Control

While most existing adaptive systems operate at single intersections, future work must focus on city-wide coordination through networked communication. Using graph-based algorithms and predictive modeling, intersections can

communicate and adapt signal timings based on real-time congestion across neighboring areas. This collaborative control can significantly reduce overall waiting time, ensure smoother vehicle progression, and enhance urban mobility efficiency.

4.2.5 Sustainable and Smart-City Integration

Future traffic systems should align with the broader vision of smart and sustainable cities. Integrating adaptive control mechanisms with public transport schedules, pollution monitoring sensors, and renewable-powered infrastructure will ensure that traffic optimization goes hand-in-hand with environmental conservation. Moreover, real-time feedback from these systems can support government authorities in making informed policy decisions related to transport planning and energy efficiency.

5. Performance Metrics

Table 1. Performance Evaluation Parameters

Metric	Definition	Formula
Precision	Measures many detected vehicles are actually correct detections.	$Precision = \frac{TP}{TP + FP}$
Recall	Measures how many actual vehicles were successfully detected.	$Recall = \frac{TP}{TP + FN}$
F1-Score	Harmonic mean of precision and recall balancing both metrics.	$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$
mAP50	Mean Average Precision calculated at IoU threshold of 0.50.	$mAP@50 = \frac{1}{N} \sum_{i=1}^N AP_i$

Table 2. YOLO Performance metrics values:

Metric	Avg. value over 100 epochs
Precision	0.9120
Recall	0.7855
F1-Score	0.8488
mAP50	0.6496

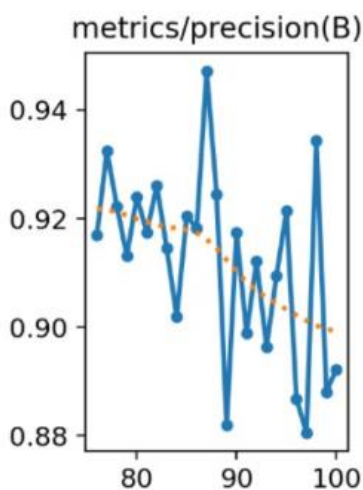


Fig.6. Precision v/s Confidence

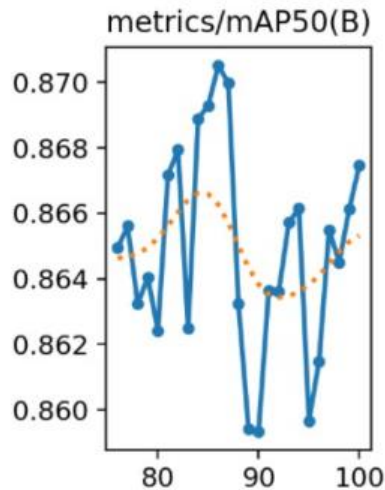


Fig.7. mAP50 v/s Confidence

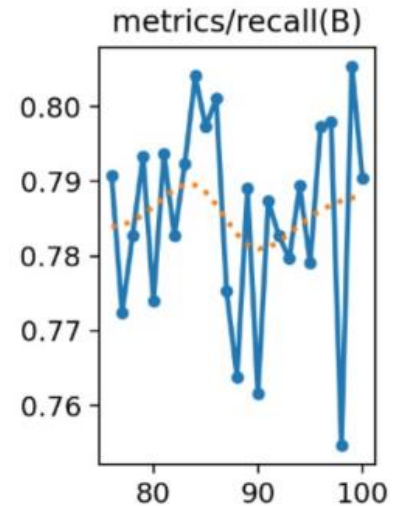


Fig.8. Recall v/s Confidence

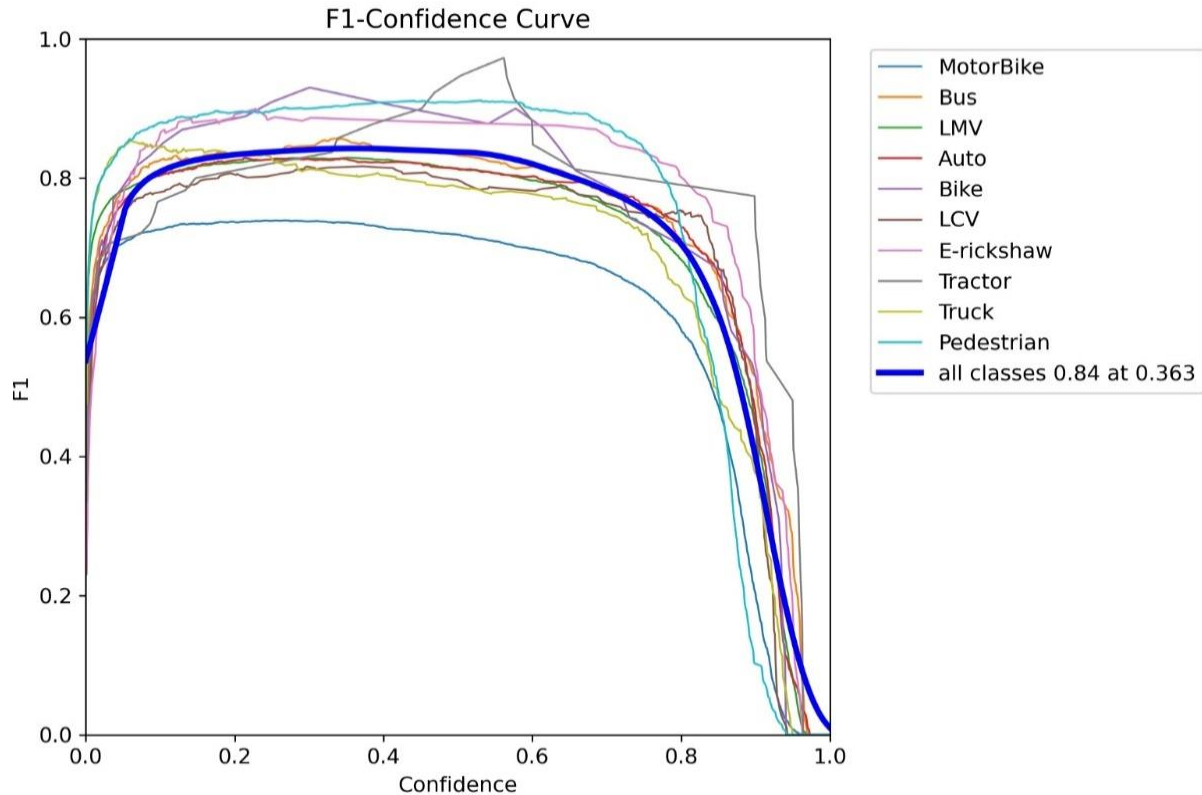


Fig.9. F1Score v/s Confidence

6. Conclusion

From this survey, it is evident that the evolution of traffic signal control has progressed from static, pre-timed systems to intelligent, vision-based adaptive frameworks. The reviewed studies show that deep learning models such as YOLO and tracking algorithms like ByteTrack have made real-time vehicle detection and motion analysis both feasible and reliable. These methods provide accurate traffic density estimation, which forms the foundation for data-driven and automated signal timing decisions.

However, most existing solutions are still limited to individual intersections and lack large-scale coordination. The literature consistently indicates that while adaptive control improves local performance, its full potential can only be realized when multiple intersections operate collaboratively through networked optimization techniques such as graph-based or reinforcement learning approaches. The studies also reveal that challenges like data scarcity, occlusion, and computational costs continue to restrict real-world deployment.

The comparative analysis across papers suggests that integrating CCTV imagery with AI-driven analytics offers a cost-effective and scalable alternative to traditional sensor-based systems. Furthermore, incorporating multimodal data sources and region-specific datasets could significantly enhance system robustness and accuracy.

The reviewed BFS-coordinated system demonstrated a 36.45% reduction in average waiting time (58.41 s $\rightarrow$ 37.12 s), 30.87% reduction in time loss, and 65.9% improvement in emergency vehicle response (51.88 s $\rightarrow$ 17.71 s). YOLOv11 detection achieved mAP@50 exceeding 0.91, confirming high precision and recall across multiple vehicle classes including cars, two-wheelers, buses, trucks, and emergency vehicles.

Based on the findings, the survey concludes that an effective adaptive traffic system should combine visual intelligence, real-time responsiveness, and multi-junction coordination. The inclusion of features such as emergency vehicle prioritization, sustainable routing, and data privacy frameworks will transform current traffic systems into

holistic smart-city solutions. Hence, the future of intelligent traffic management lies in creating unified, self-learning, and sustainable systems that can adapt dynamically to ever-changing urban mobility patterns.

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